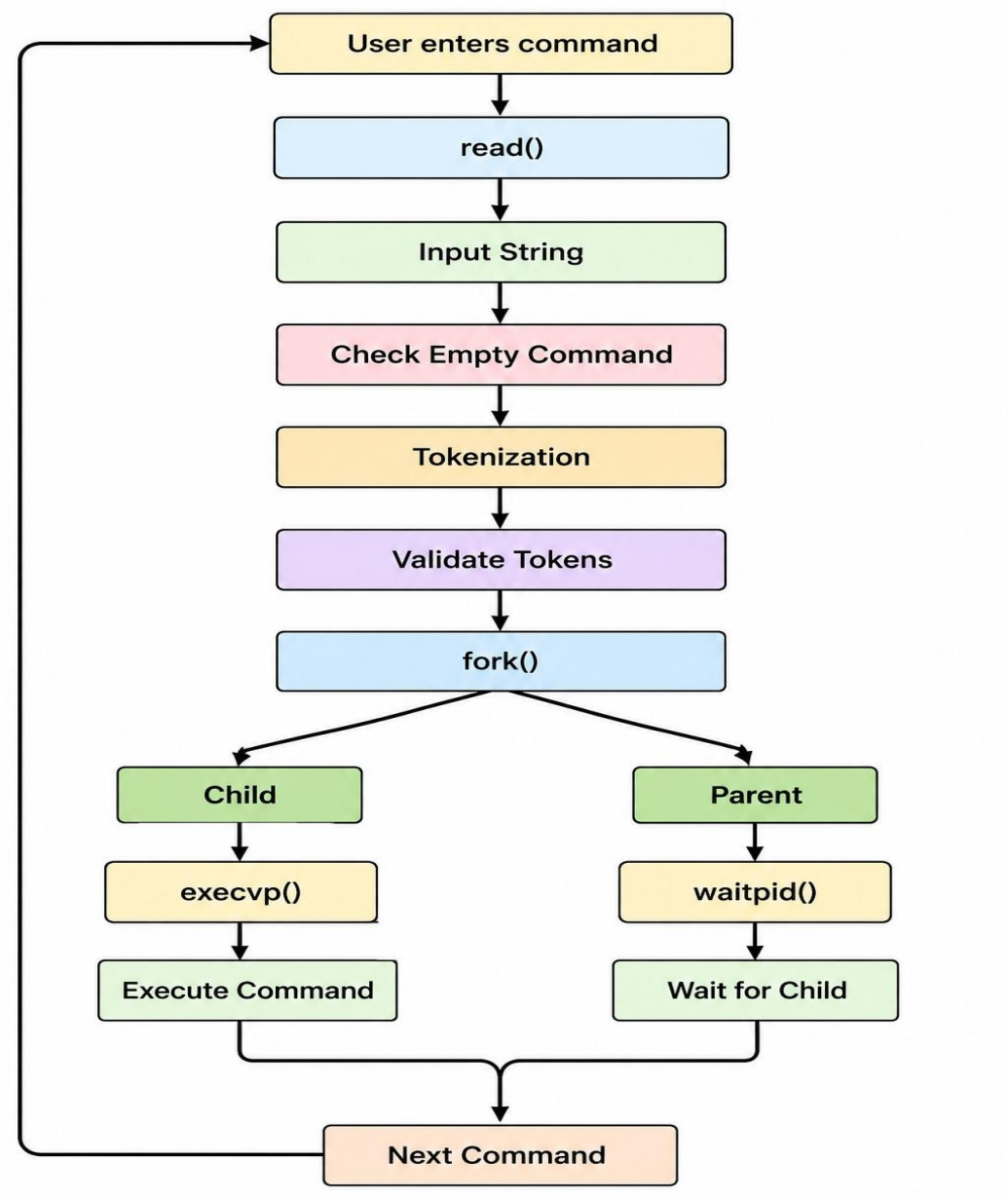
Refer the following link to understand the Parsing and Lexical analysis https://www.geeksforgeeks.org/compiler-design/introduction-of-lexical-analysis/

Task 1:

To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output and Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures Flow Chart:



CODE:

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 100

void tokenize(char input[]) {
   char *token;
   int count = 0;

   printf("\n--- TOKENS ---\n");

   token = strtok(input, " \t\n");

   while (token != NULL) {
      printf("Token %d: %s\n", ++count, token);
      token = strtok(NULL, " \t\n");
   }

   if (count == 0)
      printf("Empty command\n");
}

int main() {
   char input[MAX];

   printf("=====================================\n");
   printf("   LEXICAL ANALYSIS AND PARSING\n");
   printf("=====================================\n");

   printf("Enter command: ");
   fgets(input, MAX, stdin);

   if (strlen(input) == 1) {
      printf("\nError: Empty command.\n");
      return 0;
   }

   tokenize(input);

   printf("\n--- PARSER ---\n");
```

```
    printf("Syntax validation successful.\n");
    printf("Parse completed successfully.\n");

    printf("\n--- EXECUTION STRUCTURE ---\n");
    printf("Input Command\n");
    printf("      |\n");
    printf("   Tokenizer\n");
    printf("      |\n");
    printf("   Parser\n");
    printf("      |\n");
    printf(" Syntax Check\n");
    printf("      |\n");
    printf(" Execution Structure\n");

    return 0;
}
```

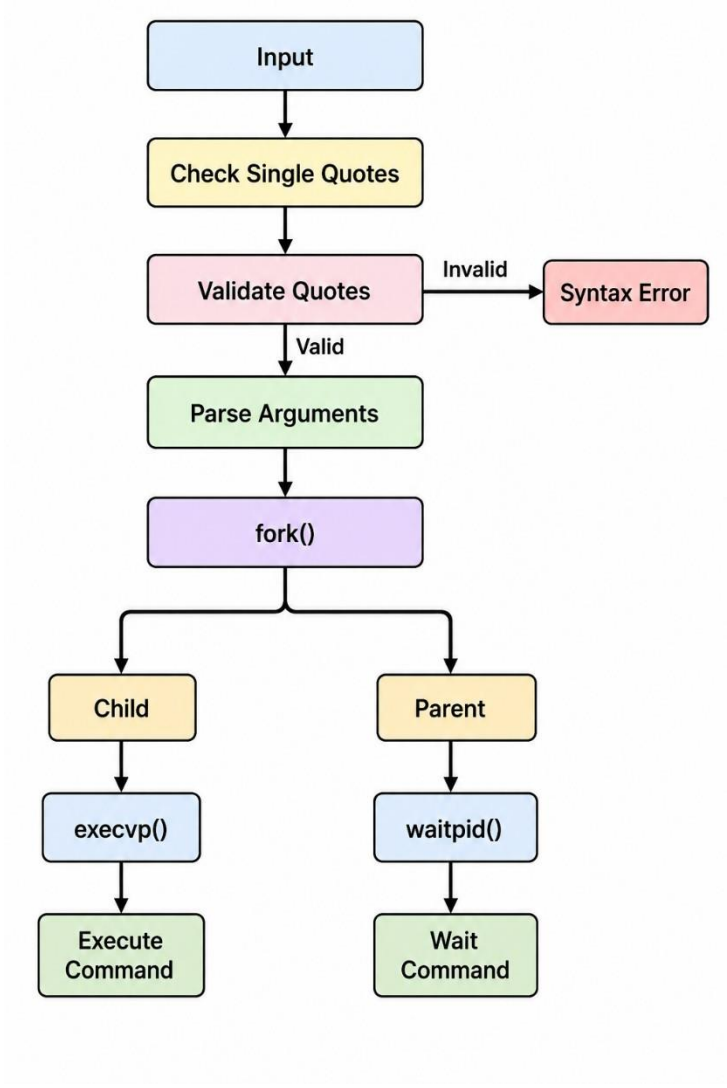
Output:

```
chakrika@Chakrika:~/OSSP$ ./task1
=====================================
   LEXICAL ANALYSIS AND PARSING
=====================================
Enter command: ls -l /home

--- TOKENS ---
Token 1: ls
Token 2: -l
Token 3: /home

--- PARSER ---
Syntax validation successful.
Parse completed successfully.

--- EXECUTION STRUCTURE ---
Input Command
      |
   Tokenizer
      |
   Parser
      |
 Syntax Check
      |
 Execution Structure
chakrika@Chakrika:~/OSSP$
```

Task 2:

To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

Flow Chart:



Example Output:
Case1:
myShell> echo Hello

Parsing Result:
Argument 1: echo
Argument 2: Hello

Child PID: 3210 Hello

Command execution completed.

Case2: myShell> echo
'Hello World

Syntax Error: Unmatched single quote.

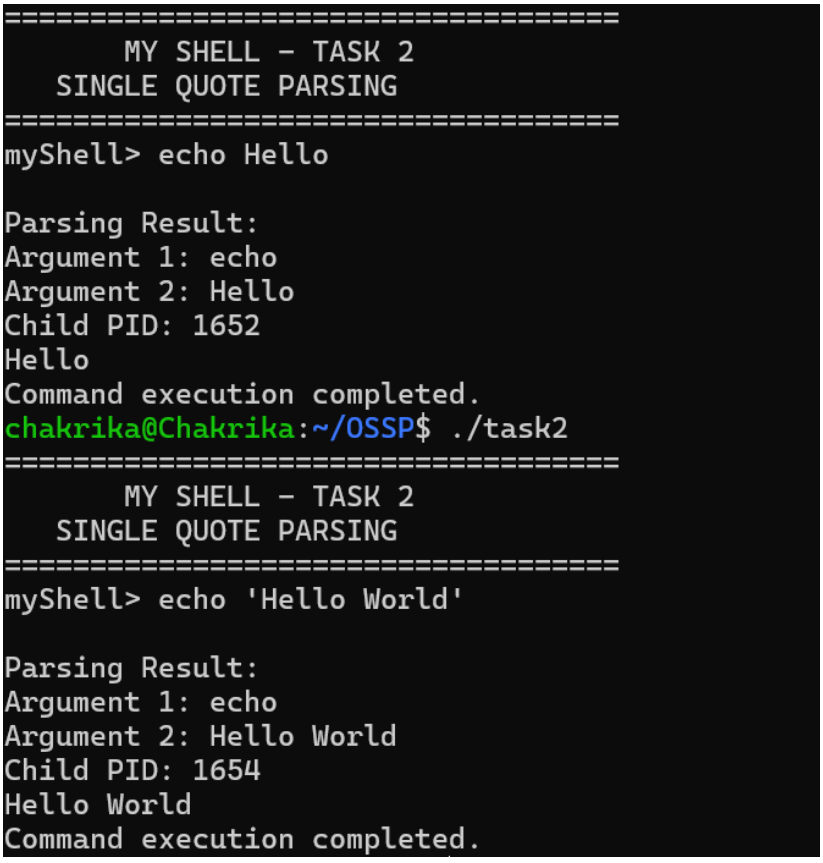
CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 256
#define MAX_ARGS 50

int parse_command(char *input, char *args[]) {
    int argc = 0, i = 0, in_quotes = 0;
    char token[MAX_INPUT];
    int j = 0;

    while (input[i] != '\0') {
        if (input[i] == '\'') {
            in_quotes = !in_quotes;
            i++;
            continue;
        }

        if ((input[i] == ' ' || input[i] == '\t' || input[i] == '\n') && !in_quotes) {
            if (j > 0) {
                token[j] = '\0';
                args[argc] = malloc(strlen(token) + 1);
                strcpy(args[argc], token);
                argc++;
                j = 0;
            }
            i++;
            continue;
        }

        token[j++] = input[i++];
    }

    if (in_quotes) {
        printf("\nSyntax Error: Unmatched single quote.\n");
        return -1;
```

```
    }

    if (j > 0) {
        token[j] = '\0';
        args[argc] = malloc(strlen(token) + 1);
        strcpy(args[argc], token);
        argc++;
    }

    args[argc] = NULL;
    return argc;
}

int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("====================================\n");
    printf("       MY SHELL - TASK 2\n");
    printf("   SINGLE QUOTE PARSING\n");
    printf("====================================\n");
    printf("myShell> ");

    if (fgets(input, sizeof(input), stdin) == NULL) {
        return 1;
    }

    int argc = parse_command(input, args);

    if (argc == -1) {
        return 0;
    }

    if (argc == 0) {
        printf("\nSyntax Error: Empty command.\n");
        return 0;
    }

    printf("\nParsing Result:\n");

    for (int i = 0; i < argc; i++) {
        printf("Argument %d: %s\n", i + 1, args[i]);
    }

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
```

```
    }

    if (pid == 0) {
        printf("Child PID: %d\n", getpid());
        execvp(args[0], args);
        perror("execvp");
        exit(1);
    } else {
        int status;
        waitpid(pid, &status, 0);
        printf("Command execution completed.\n");
    }

    for (int i = 0; i < argc; i++) {
        free(args[i]);
    }

    return 0;
}
```

Output:



Task 3:

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

Example Output:

Case1: myShell> echo "Hello
$USER"

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello raghu

Child Process PID
= 3022

Hello raghu

Child process completed.
Case2: myShell> echo

"Hello World

Syntax Error: Unmatched double quote.

CODE:

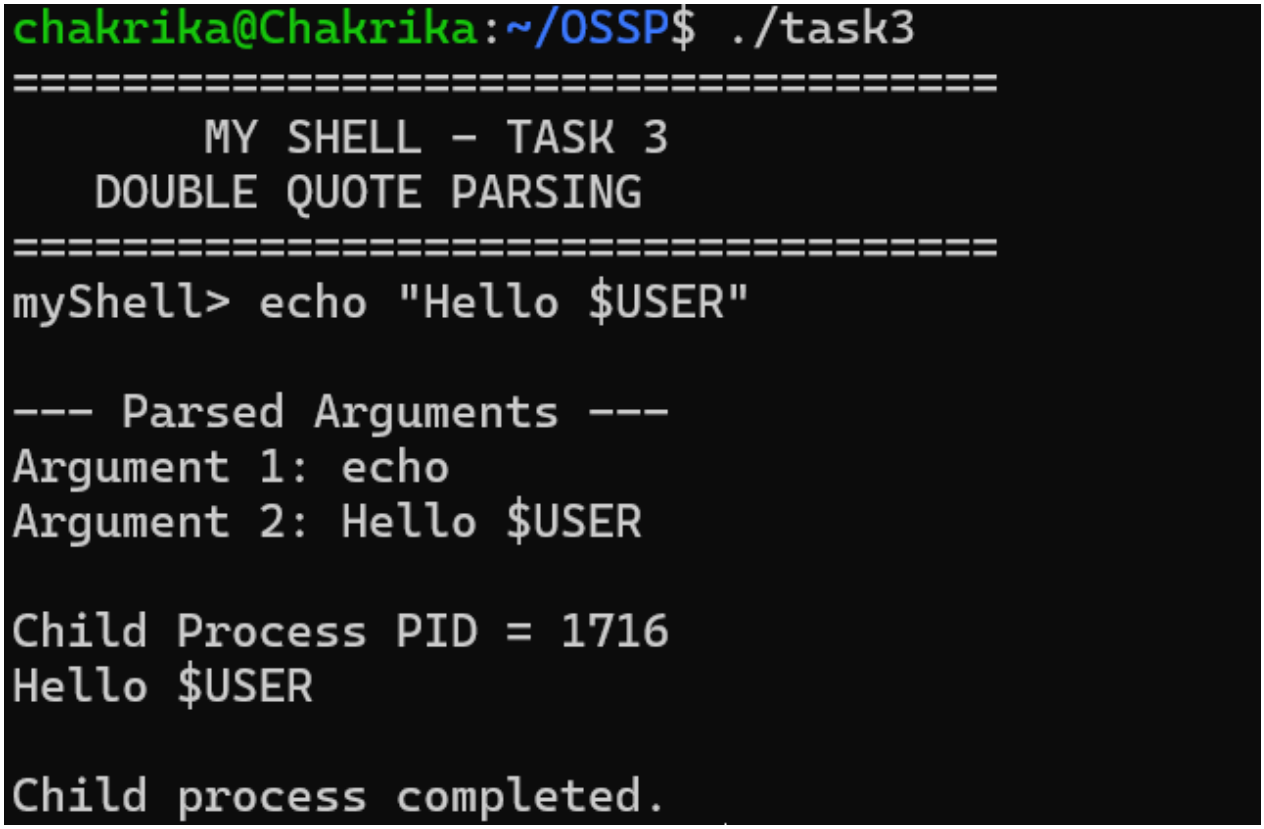
```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 256
#define MAX_ARGS 50
```

```
int parse_command(char *input, char *args[]) {
    int argc = 0, i = 0, in_quotes = 0;
    char quote_type = 0;
    char token[MAX_INPUT];
    int j = 0;

    while (input[i] != '\0') {
        if (input[i] == '"') {
            if (!in_quotes) {
                in_quotes = 1;
                quote_type = '"';
            } else if (quote_type == '"') {
                in_quotes = 0;
                quote_type = 0;
            }
            i++;
            continue;
        }

        if ((input[i] == ' ' || input[i] == '\t' || input[i] == '\n') && !in_quotes) {
            if (j > 0) {
                token[j] = '\0';
                args[argc] = malloc(strlen(token) + 1);
                strcpy(args[argc], token);
                argc++;
                j = 0;
            }
            i++;
            continue;
        }

        token[j++] = input[i++];
    }

    if (in_quotes) {
        printf("\nSyntax Error: Unmatched double quote.\n");
        return -1;
    }

    if (j > 0) {
        token[j] = '\0';
        args[argc] = malloc(strlen(token) + 1);
        strcpy(args[argc], token);
        argc++;
    }

    args[argc] = NULL;
    return argc;
}
```

```
int main() {
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("====================================\n");
    printf("       MY SHELL - TASK 3\n");
    printf("   DOUBLE QUOTE PARSING\n");
    printf("====================================\n");
    printf("myShell> ");

    if (fgets(input, sizeof(input), stdin) == NULL)
        return 1;

    int argc = parse_command(input, args);

    if (argc == -1)
        return 0;

    if (argc == 0) {
        printf("\nSyntax Error: Empty command.\n");
        return 0;
    }

    printf("\n--- Parsed Arguments ---\n");

    for (int i = 0; i < argc; i++)
        printf("Argument %d: %s\n", i + 1, args[i]);

    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process PID = %d\n", getpid());

        execvp(args[0], args);

        perror("execvp");
        exit(1);
    } else {
        int status;
        waitpid(pid, &status, 0);
        printf("\nChild process completed.\n");
    }
```

```
    for (int i = 0; i < argc; i++)
        free(args[i]);

    return 0;
}
```

OUTPUT:



Task 4: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/wait.h>

#define MAX_INPUT 256
```

```
#define MAX_ARGS 50

int parse_command(char *input, char *args[]) {
    int argc = 0, i = 0;
    int escaped = 0;
    char token[MAX_INPUT];
    int j = 0;

    while (input[i] != '\0') {
        if (escaped) {
            token[j++] = input[i];
            escaped = 0;
            i++;
            continue;
        }

        if (input[i] == '\\') {
            escaped = 1;
            i++;
            continue;
        }

        if (input[i] == ' ' || input[i] == '\t' || input[i] == '\n') {
            if (j > 0) {
                token[j] = '\0';
                args[argc] = malloc(strlen(token) + 1);
                strcpy(args[argc], token);
                argc++;
                j = 0;
            }
            i++;
            continue;
        }

        token[j++] = input[i++];
    }

    if (escaped) {
```

```
            token[j++] = '\\';
        }

        if (j > 0) {
            token[j] = '\0';
            args[argc] = malloc(strlen(token) + 1);
            strcpy(args[argc], token);
            argc++;
        }

        args[argc] = NULL;
        return argc;
    }

    int main() {
        char input[MAX_INPUT];
        char *args[MAX_ARGS];

        printf("=====================================\n");
        printf("       MY SHELL - TASK 4\n");
        printf("   ESCAPE SEQUENCE PARSING\n");
        printf("=====================================\n");
        printf("myShell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)
            return 1;

        int argc = parse_command(input, args);

        if (argc == 0) {
            printf("\nSyntax Error: Empty command.\n");
            return 0;
        }

        printf("\n--- Parsed Arguments ---\n");

        for (int i = 0; i < argc; i++)
            printf("Argument %d: %s\n", i + 1, args[i]);
```

```
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process PID = %d\n", getpid());

        execvp(args[0], args);

        perror("execvp");
        exit(1);
    } else {
        int status;
        waitpid(pid, &status, 0);

        printf("\nChild process completed.\n");
    }

    for (int i = 0; i < argc; i++)
        free(args[i]);

    return 0;
}
```

OUTPUT:

CASE 1:

```
==================================
       MY SHELL - TASK 4
   ESCAPE SEQUENCE PARSING
==================================
myShell> echo Hello\ World

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello World

Child Process PID = 1781
Hello World

Child process completed.

Child process completed.
```

CASE 2:

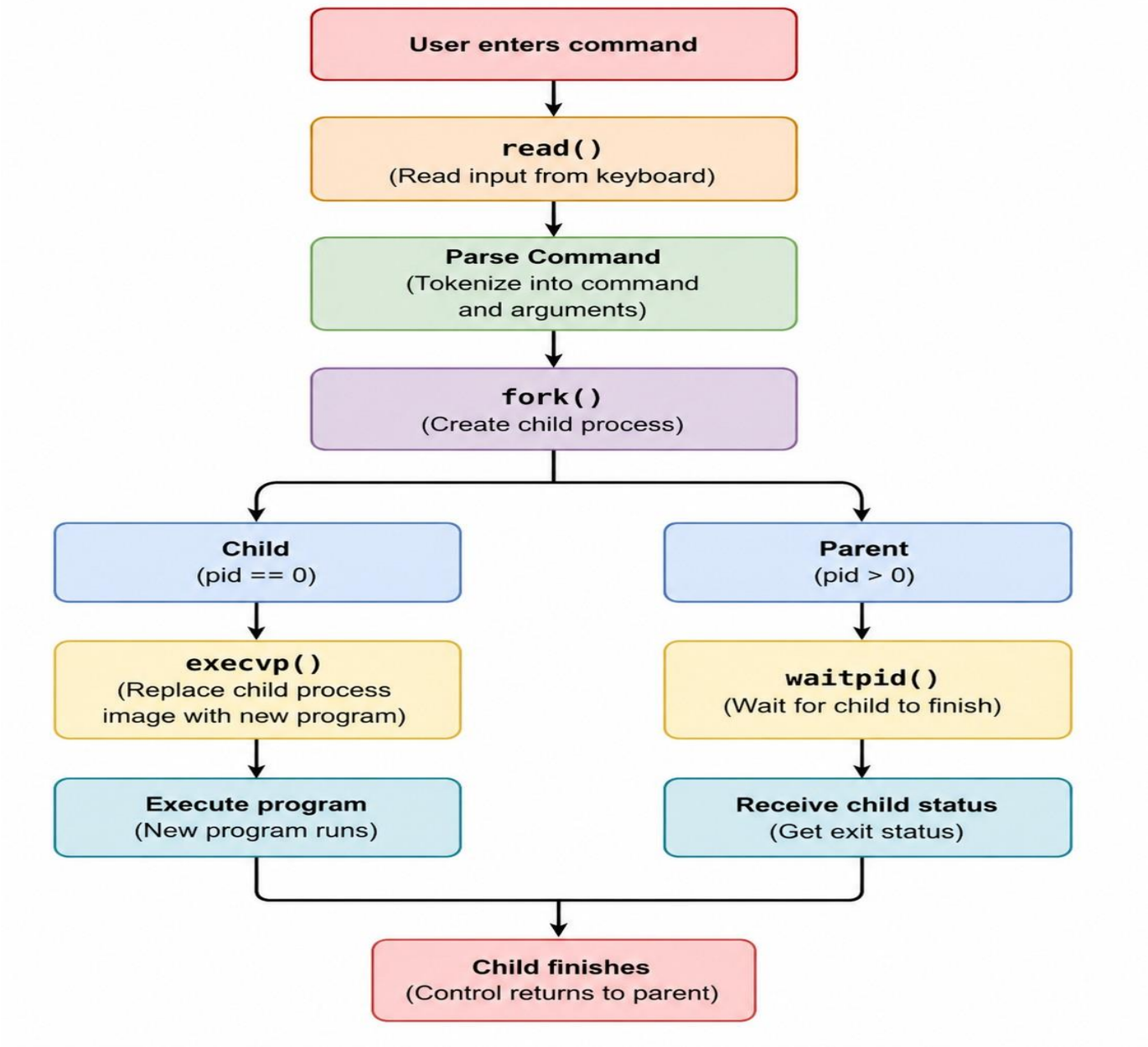
```
chakrika@Chakrika:~/OSSP$ ./task4
==================================
       MY SHELL - TASK 4
   ESCAPE SEQUENCE PARSING
==================================
myShell> echo Hello\&World

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello&World

Child Process PID = 1813
Hello&World

Child process completed.
```

CASE 3:

```
chakrika@Chakrika:~/OSSP$ ./task4
==================================
       MY SHELL - TASK 4
   ESCAPE SEQUENCE PARSING
==================================
myShell> echo Hello\ World\ \&\ Test

--- Parsed Arguments ---
Argument 1: echo
Argument 2: Hello World & Test

Child Process PID = 1821
Hello World & Test

Child process completed.
```

Task 5: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching.

Flow Chart:



Sample Output:
Case 1: myshell$
ls
Parent Process PID: 1200
Created Child PID: 1201
Child Process PID: 1201
file1.txt file2.txt
Child exited with status: 0

Case 2:
myshell$ ls @l Parent
Process PID: 82
Child Process PID: 83 Created
Child PID: 83
ls: cannot access '@l': No such file or directory
Child exited with status: 2

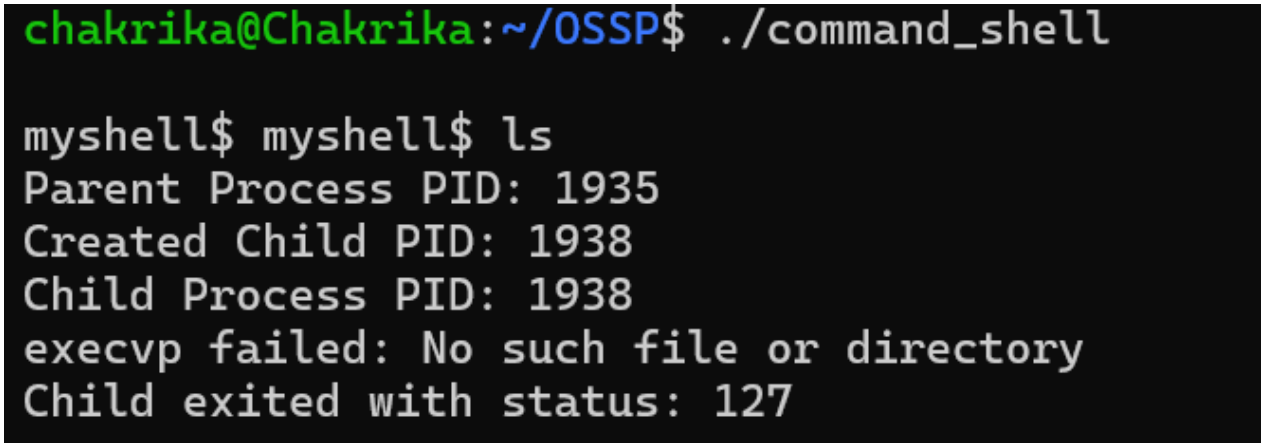
CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_INPUT 100
#define MAX_ARGS 20

int main() {
   char input[MAX_INPUT];

   while (1) {
      printf("\nmyshell$ ");
      fflush(stdout);

      if (fgets(input, sizeof(input), stdin) == NULL)
         break;
```

```
input[strcspn(input, "\n")] = '\0';

if (strlen(input) == 0)
    continue;

if (strcmp(input, "exit") == 0)
    break;

char *args[MAX_ARGS];
int i = 0;

char *token = strtok(input, " ");

while (token != NULL && i < MAX_ARGS - 1) {
    args[i++] = token;
    token = strtok(NULL, " ");
}

args[i] = NULL;

printf("Parent Process PID: %d\n", getpid());

pid_t pid = fork();

if (pid < 0) {
    perror("fork failed");
    continue;
}

if (pid == 0) {
    printf("Child Process PID: %d\n", getpid());

    execvp(args[0], args);

    perror("execvp failed");
    exit(127);
```

```
        }
        else {
            printf("Created Child PID: %d\n", pid);

            int status;
            waitpid(pid, &status, 0);

            if (WIFEXITED(status)) {
                printf("Child exited with status: %d\n",
                    WEXITSTATUS(status));
            }
        }
    }

    return 0;
}
```

OUTPUT:
CASE 1:



CASE 2:

```
myshell$ myshell$ ls @l
Parent Process PID: 1935
Created Child PID: 1943
Child Process PID: 1943
execvp failed: No such file or directory
Child exited with status: 127

myshell$ myshell$ exit
Parent Process PID: 1935
Created Child PID: 1944
Child Process PID: 1944
execvp failed: No such file or directory
Child exited with status: 127
```